

MASTER'S COMPREHENSIVE EXAMINATION

Part I Theory (2 hours) March 28th, 2015

Instructions: Attempt any three out of the following four questions on both pages. Mention which three you would like to have graded. Open book and open notes.

1. Suppose that $\mathbf{X}$ is a random vector with a multivariate normal distribution such that $\Sigma^{-1/2}(\mathbf{X} - \boldsymbol{\mu})$ has zero mean and variance covariance matrix equal to the identity ($N(0, I)$). Suppose we observe a sample $\mathbf{Y}_1, \dots, \mathbf{Y}_n$ from the distribution of the vector $\mathbf{X} = (X_1, X_2, \dots, X_p)^T$ and suppose that $\boldsymbol{\mu} = (\mu_1, \mu_2, \dots, \mu_p)^T$ and $\Sigma = (\sigma_{ij})$ $i = 1, \dots, p, j = 1, \dots, p$.

Find the method of moments (MOM) estimators of $\mu_1, \mu_2, \dots, \mu_p$ and σ_{ij} $i \leq j$.

2. Let X be a sample of size=1 from a Poisson distribution with parameter λ , and let $h(\lambda) = e^{-3\lambda}$.

- (a) Check if $T = (-2)^X$ is an unbiased estimator of $h(\lambda)$.
(b) Give some bad property of the above estimator, which is unrelated to the question of bias.

3. My first serve percentage for tennis was 42 percent good serves. I took lessons and afterwards tried 100 serves. I got 48 good serves. Did I improve? (Test at $\alpha = 0.05$).

4. Let $Y_1, \dots, Y_n$ be i.i.d. normally distributed with mean μ , and standard deviation σ .

a) Find the maximum likelihood estimate of σ .

b) Find the maximum likelihood estimate of $\mu + 1.645 \sigma$ (that is the 95th percentile of the distribution of Y).